REVIEW PACKAGE — STOP BEFORE MANUFACTURING EXPORTS

USB-powered Teensy / six socketed Minis / JST SMD + hand THT / fused 12V motor power

Identical global labels are the same connection. Hardware kill buffers gate all six EN lines.

Drill-verified Mini v1.0 footprint; unchanged encoder mapping; 3+3 placement, 150 x 140 mm. Review before ordering.

01 / USB-powered Teensy sockets

File: controller.kicad_sch

02 / A1: index_curl_1

File: axis1.kicad_sch

03 / A2: index_curl_2

File: axis2.kicad_sch

04 / A3: index_splay

File: axis3.kicad_sch

05 / A4: middle_curl_1

File: axis4.kicad_sch

06 / A5: middle_curl_2

File: axis5.kicad_sch

07 / A6: middle_splay

File: axis6.kicad_sch

08 / Power and hardware kill

File: power.kicad_sch

09 / Fused reverse-polarity protection

File: protection.kicad_sch

Sheet: /
File: trueskate-main-v1.kicad_sch

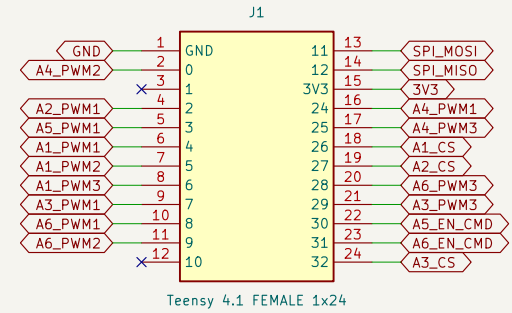
Title: TrueSkate USB-powered six-axis carrier

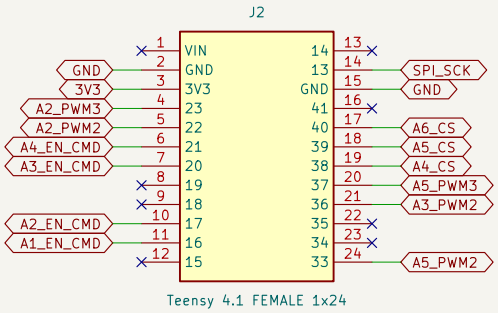
Size: A3	Date:	**Rev: v1 REVIEW**
KiCad E.D.A. 10.0.6		Id: 1/10

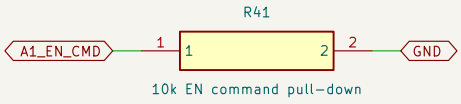
USB is the ONLY Teensy power source. VIN socket contact is deliberately unconnected.

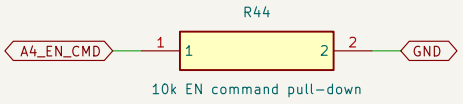
Pin map unchanged. EN commands pass through hardware kill buffers on the power sheet.

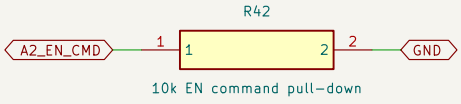
Teensy 3.3V powers encoders and buffers. Mini regulator outputs remain isolated.

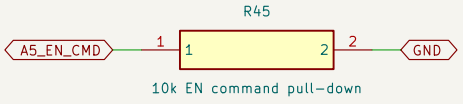


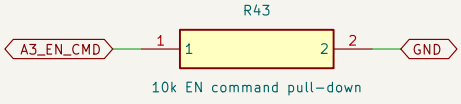


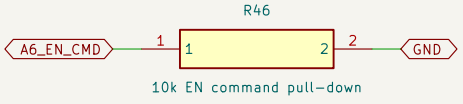










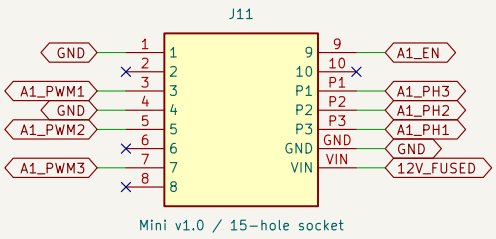


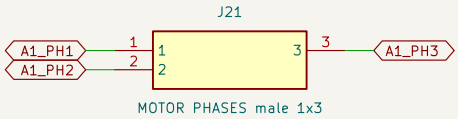
Sheet: /01 / USB-powered Teensy sockets/
File: controller.kicad_sch

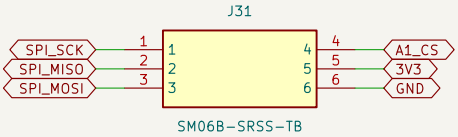
Title: 01 / USB-powered Teensy sockets

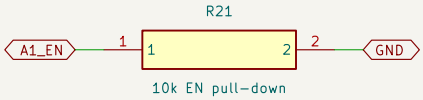
Size: A4	Date:	**Rev: v1 REVIEW**
KiCad E.D.A. 10.0.6		Id: 2/10

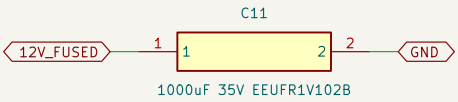
Entire axis population optional. No shared connection depends on a fitted axis component.
Mini v1.0 exact 15-hole footprint, NOT mirrored. Buffered EN with local 10k pull-down.
SH order UNVERIFIED: VERIFY WITH MULTIMETER AGAINST AS5048A BOARD. See README table.
Pin labels derive from design.json encoder_pin_order. CS has an individual 10k pull-up.

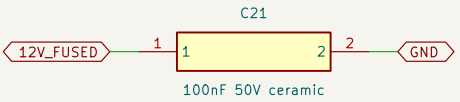


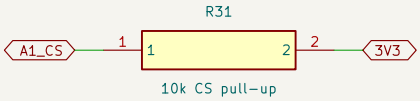






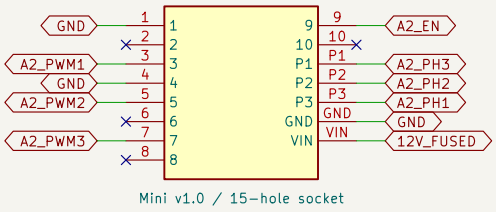


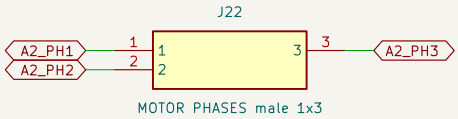


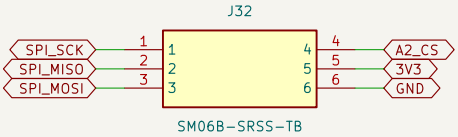


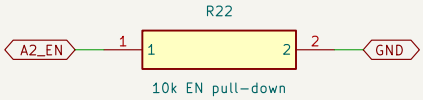
Sheet: /02 / A1: index_curl_1/		
File: axis1.kicad_sch		
Title: 02 / A1: index_curl_1		
Size: A4	Date:	**Rev: v1 REVIEW**
KiCad E.D.A. 10.0.6		Id: 3/10

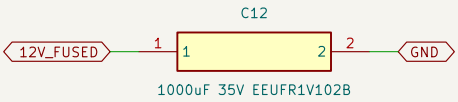
Entire axis population optional. No shared connection depends on a fitted axis component.
Mini v1.0 exact 15-hole footprint, NOT mirrored. Buffered EN with local 10k pull-down.
SH order UNVERIFIED: VERIFY WITH MULTIMETER AGAINST AS5048A BOARD. See README table.
Pin labels derive from design.json encoder_pin_order. CS has an individual 10k pull-up.

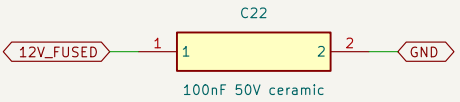


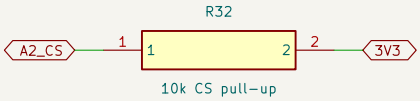










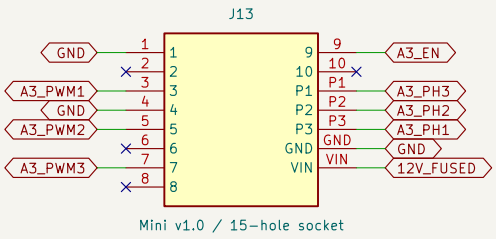


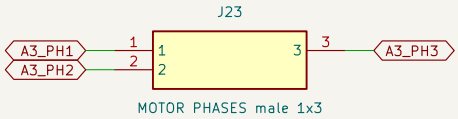
Sheet: /03 / A2: index_curl_2/
File: axis2.kicad_sch

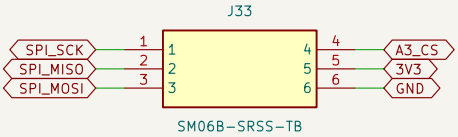
Title: 03 / A2: index_curl_2

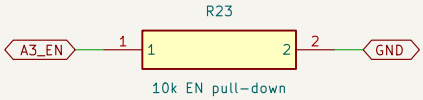
Size: A4	Date:	**Rev: v1 REVIEW**
KiCad E.D.A. 10.0.6		Id: 4/10

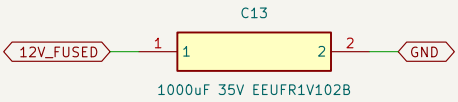
Entire axis population optional. No shared connection depends on a fitted axis component.
Mini v1.0 exact 15-hole footprint, NOT mirrored. Buffered EN with local 10k pull-down.
SH order UNVERIFIED: VERIFY WITH MULTIMETER AGAINST AS5048A BOARD. See README table.
Pin labels derive from design.json encoder_pin_order. CS has an individual 10k pull-up.

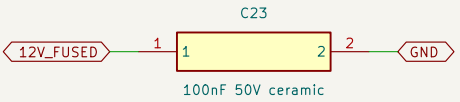


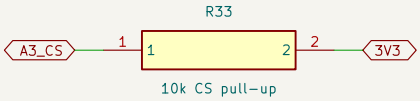










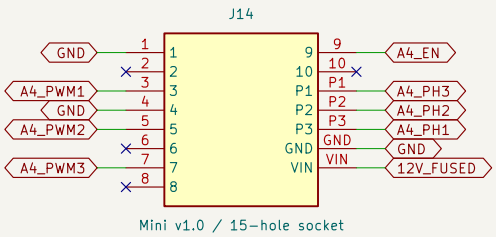


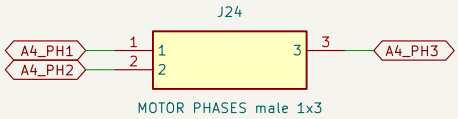
Sheet: /04 / A3: index_splay/
File: axis3.kicad_sch

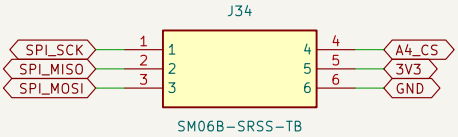
Title: 04 / A3: index_splay

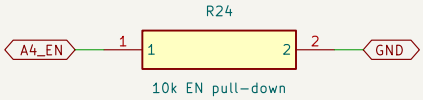
Size: A4	Date:	Rev: v1 REVIEW
KiCad E.D.A. 10.0.6		Id: 5/10

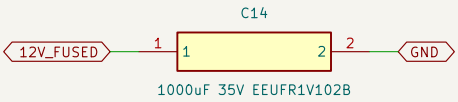
Entire axis population optional. No shared connection depends on a fitted axis component.
Mini v1.0 exact 15-hole footprint, NOT mirrored. Buffered EN with local 10k pull-down.
SH order UNVERIFIED: VERIFY WITH MULTIMETER AGAINST AS5048A BOARD. See README table.
Pin labels derive from design.json encoder_pin_order. CS has an individual 10k pull-up.

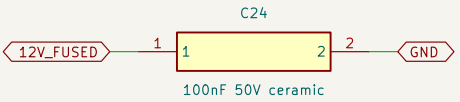


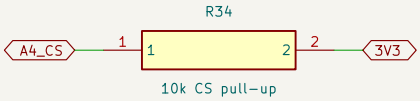






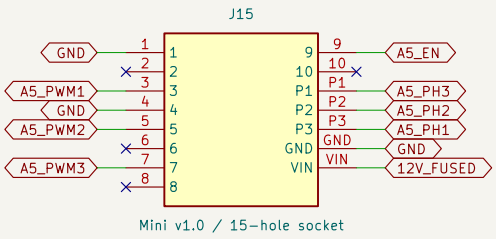


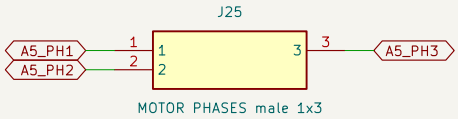


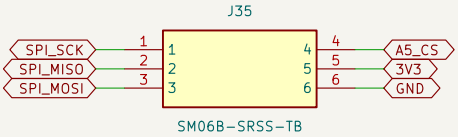


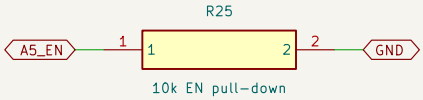
Sheet: /05 / A4: middle_curl_1/		
File: axis4.kicad_sch		
Title: 05 / A4: middle_curl_1		
Size: A4	Date:	**Rev: v1 REVIEW**
KiCad E.D.A. 10.0.6		Id: 6/10

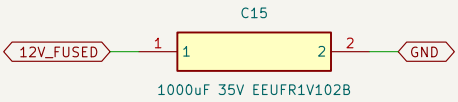
Entire axis population optional. No shared connection depends on a fitted axis component.
Mini v1.0 exact 15-hole footprint, NOT mirrored. Buffered EN with local 10k pull-down.
SH order UNVERIFIED: VERIFY WITH MULTIMETER AGAINST AS5048A BOARD. See README table.
Pin labels derive from design.json encoder_pin_order. CS has an individual 10k pull-up.

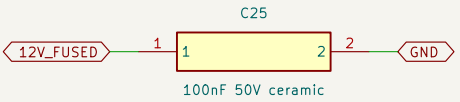


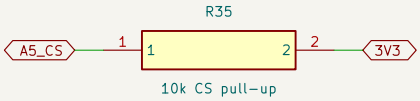






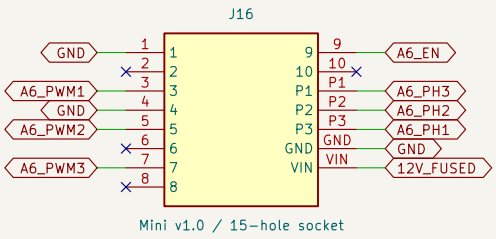


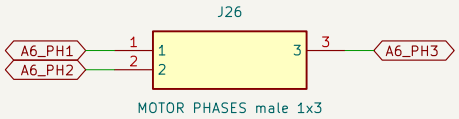


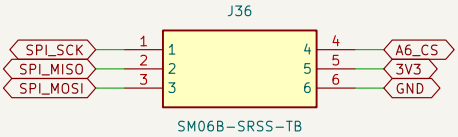


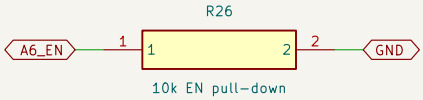
Sheet: /06 / A5: middle_curl_2/		
File: axis5.kicad_sch		
Title: 06 / A5: middle_curl_2		
Size: A4	Date:	**Rev: v1 REVIEW**
KiCad E.D.A. 10.0.6		Id: 7/10

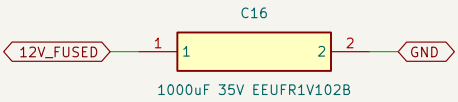
Entire axis population optional. No shared connection depends on a fitted axis component.
Mini v1.0 exact 15-hole footprint, NOT mirrored. Buffered EN with local 10k pull-down.
SH order UNVERIFIED: VERIFY WITH MULTIMETER AGAINST AS5048A BOARD. See README table.
Pin labels derive from design.json encoder_pin_order. CS has an individual 10k pull-up.

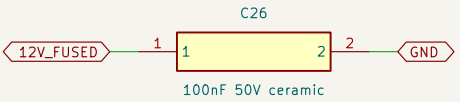


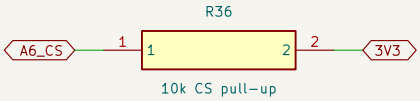












Sheet: /07 / A6: middle_splay/
File: axis6.kicad_sch

Title: 07 / A6: middle_splay

Size: A4	Date:	Rev: v1 REVIEW
KiCad E.D.A. 10.0.6		Id: 8/10

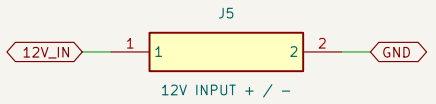
12V motor rail only. 12mm shared bus; 2mm per-axis feeds; 1oz copper. Thermal evaluation stopped.

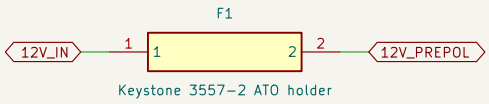
NC switch across J6: closed permits EN commands, open tri-states every EN output.

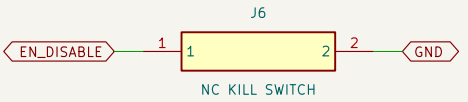
R1 pulls /OE high on open loop. Local R21-R26 pull EN low independently of firmware.

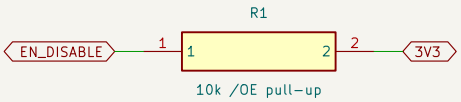
Reclosing switch restores commanded state; no restart latch. Kill does NOT isolate 12V.

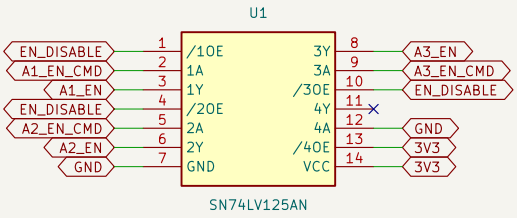
U1/U2 SN74LV125AN PDIP14, 3.3V, Ioff. Unused inputs tied; unused outputs NC.

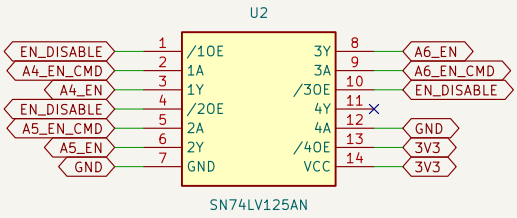


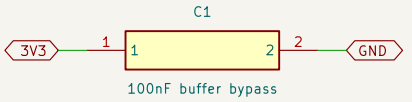


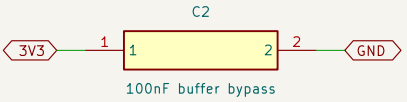










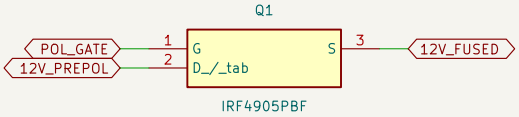


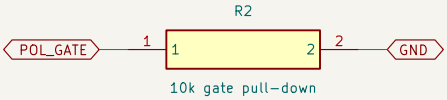
Sheet: /08 / Power and hardware kill/
File: power.kicad_sch

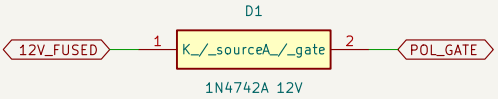
Title: 08 / Power and hardware kill

Size: A4	Date:	Rev: v1 REVIEW
KiCad E.D.A. 10.0.6		Id: 9/10

F1 -> Q1 drain; Q1 source -> protected 12V_FUSED bus. P-MOS body diode points input to load.
D1 cathode (band/pad1) to source; anode to gate. R2 10k from gate to GND.
IRF4905PbF: -55V, 20mOhm max at VGS=-10V, -74A case rating. TO-220 pin1 G,2 D/tab,3 S.
At 12A: 2.88W at 20mOhm, more when hot. Heat sink / assembly thermal verification required.
Single MOSFET is reverse-input protection, not reverse-current or load-disconnect protection.







Sheet: /09 / Fused reverse-polarity protection/
File: protection.kicad_sch

Title: 09 / Fused reverse-polarity protection

Size: A4	Date:	**Rev: v1 REVIEW**
KiCad E.D.A. 10.0.6		Id: 10/10